Discovery of Parkinson's disease states and disease progression modelling: a longitudinal data study using machine learning

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Lancet Digit Health 2021; 3: e555–64

Published Online July 29, 2021 https://doi.org/10.1016/S2589-7500(21)00101-1

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Summary

Background Parkinson's disease is heterogeneous in symptom presentation and progression. Increased understanding of both aspects can enable better patient management and improve clinical trial design. Previous approaches to modelling Parkinson's disease progression assumed static progression trajectories within subgroups and have not adequately accounted for complex medication effects. Our objective was to develop a statistical progression model of Parkinson's disease that accounts for intra-individual and inter-individual variability and medication effects.

Methods In this longitudinal data study, data were collected for up to 7-years on 423 patients with early Parkinson's disease and 196 healthy controls from the Parkinson's Progression Markers Initiative (PPMI) longitudinal observational study. A contrastive latent variable model was applied followed by a novel personalised input-output hidden Markov model to define disease states. Clinical significance of the states was assessed using statistical tests on seven key motor or cognitive outcomes (mild cognitive impairment, dementia, dyskinesia, presence of motor fluctuations, functional impairment from motor fluctuations, Hoehn and Yahr score, and death) not used in the learning phase. The results were validated in an independent sample of 610 patients with Parkinson's disease from the National Institute of Neurological Disorders and Stroke Parkinson's Disease Biomarker Program (PDBP).

Findings PPMI data were download July 25, 2018, medication information was downloaded on Sept 24, 2018, and PDBP data were downloaded between June 15 and June 24, 2020. The model discovered eight disease states, which are primarily differentiated by functional impairment, tremor, bradykinesia, and neuropsychiatric measures. State 8, the terminal state, had the highest prevalence of key clinical outcomes including 18 (95%) of 19 recorded instances of dementia. At study outset 4 (1%) of 333 patients were in state 8 and 138 (41%) of 333 patients reached stage 8 by year 5. However, the ranking of the starting state did not match the ranking of reaching state 8 within 5 years. Overall, patients starting in state 5 had the shortest time to terminal state (median 2·75 [95% CI 1·75–4·25] years).

Interpretation We developed a statistical progression model of early Parkinson's disease that accounts for intra-individual and inter-individual variability and medication effects. Our predictive model discovered non-sequential, overlapping disease progression trajectories, supporting the use of non-deterministic disease progression models, and suggesting static subtype assignment might be ineffective at capturing the full spectrum of Parkinson's disease progression.

Funding Michael J Fox Foundation.

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Introduction

Parkinson's disease is a progressive, multisystem neurodegenerative disorder with a complex pathophysiology. Bradykinesia, rigidity, rest tremor, and postural instability constitute the core motor syndrome,[1] but various other motor and non-motor signs and symptoms occur. The trajectory of patients with Parkinson's disease over time is highly variable, with some patients following a relatively benign course, and others progressing rapidly to disability.[2] The resulting inter-individual and intra-individual variability in Parkinson's disease expression in turn leads to inaccuracy of diagnosis, especially early in disease,[3] and necessitates a highly individualised approach to management and prognostication.

Recognising the heterogeneity of Parkinson's disease, it has been proposed that there are disease subtypes and there have been attempts to define these based on expert opinion[4] and via machine learning approaches.[5,6] Past work has focused on identifying subtypes of Parkinson's disease primarily using cross-sectional data under the assumption that subtype assignment at baseline is indicative of disease progression, and indeed some of the discovered subtypes have been associated with varying rates of disease progression.[7–11] In this work, we instead propose disease states, each of which is associated with particular symptom manifestations and state progression patterns, which are learned using longitudinal data. Being able to accurately define a

Research in context

Evidence before this study
Parkinson's disease is a neurodegenerative disorder heterogeneous in both its clinical manifestations and progression, which serves as possible evidence for the existence of disease subtypes. We searched PubMed and Google Scholar for original articles published in English with the terms "Parkinson's disease" and "subtypes" or "clusters" published up until Aug 15, 2020. Previous studies have identified disease subtypes using data-driven approaches, but replication across cohorts has not always been possible, and translation of these subtypes into clinical practice has not yet been achieved. Models developed to date have several key limitations including use of cross-sectional data to define the subtypes, model assumptions that each subtype follows a fixed progression, and not accounting for both positive and negative effects of symptomatic therapies for Parkinson's disease. Allowing for heterogeneity not only in disease manifestations but also progression, and accounting for medication effects is critical towards developing accurate disease models that can be used in clinical and research settings.

Added value of this study
In this study, we address the limitations of previous work. We relax the progression constraints imposed in previous work by proposing a machine learning model that allows patients to possibly follow different progression pathways among disease states. This additional flexibility is coupled with modelling that accounts for medication effects and population-level variation. Combined, these modelling decisions allow many possible progression patterns while still using a small number of states to describe patient status. The parameters of the model are learned using longitudinal data from a multicentre, international, observational cohort study of 423 individuals with early Parkinson's disease, untreated at baseline, and a comparator group of 196 healthy controls. These modelling decisions enable us to learn more descriptive disease states and progression pathways as shown via association with key clinical outcomes. The results are validated in an independent multicentre Parkinson's disease cohort dataset comprised of 610 individuals.

Implications of all the available evidence
This study presents a statistical progression model of early Parkinson's disease that accounts for intra-individual and inter-individual variability and medication effects, circumventing limitations of previous models that have not accounted for these important factors. Using data collected over up to 7 years, we identified eight unique states defined by relative differences in motor or non-motor Parkinson's disease manifestations. Although the states were constrained to be progressive, sequential progression through disease states was not observed. Thus, our model discovered non-sequential, overlapping disease progression trajectories, supporting the use of non-deterministic disease progression models, and suggesting static subtype assignment might be ineffective at capturing the full spectrum of Parkinson's disease progression. This finding supports the hypothesis of heterogeneous progression and has implications for patient stratification and management. Finally, our work is an example of the benefits of machine learning, which can be used to translate the complexity of a chronic disease such as Parkinson's disease into a useful and interpretable model.

patient's state early in the disease course may have clinical applications, such as prognostication and patient counselling, as well as applications in clinical trial design, including subject selection and sample enrichment.[12] However, a crucial factor to consider in defining Parkinson's disease states is the effect Parkinson's disease medications have on disease measures. Most patients in observational studies initiate symptomatic therapy within 2 years of diagnosis.[13] Response to symptomatic therapy, as well as side-effects, both heavily influence an individual patient's clinical presentation at any given time, as well as measures of their trajectory longitudinally. Failure to account for these effects, especially in early Parkinson's disease, risks yielding inaccurate models. To address this critical gap, the objective of this analysis was to develop a statistical progression model that could be used to gain insights about the progression of Parkinson's disease while accounting for intra-individual and inter-individual variability and medication effects, circumventing limitations of previous attempts that have not accounted for these factors.

See Online for appendix

For more on the **PPMI study methodology** see https://www.ppmi-info.org/

Methods

Study design and participants
This longitudinal data study used data from the Parkinson's Progression Markers Initiative (PPMI), a multicentre longitudinal observational study of a highly characterised cohort, and applied a contrastive latent variable model followed by a novel hidden Markov model variant (figure 1). The resulting model was validated on a held-out test cohort from PPMI and on an independent sample of patients with Parkinson's disease enrolled in the National Institute of Neurological Disorders and Stroke Parkinson's Disease Biomarker Program (PDBP). Methods are described in the following paragraphs, with additional details in the appendix (pp 4–8).

PPMI study methodology has been described elsewhere in detail[14] and are available on the PPMI website. PPMI recruited individuals with early Parkinson's disease diagnosed within 2 years of enrollment across 24 study sites.[15] Participants with Parkinson's disease were untreated at baseline, not expected to require symptomatic therapy for 6 months, and had dopamine transporter deficit on single-photon-emission CT

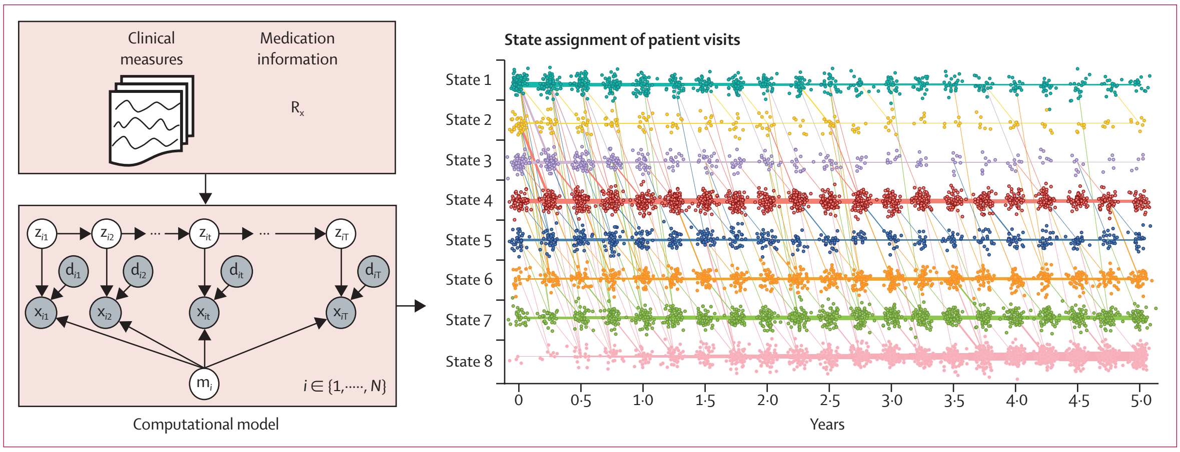


Figure 1: **Research overview**
A computational model, specifically a personalised input-output hidden Markov model, of Parkinson's disease progression is learned and applied using longitudinal clinical and imaging measures and medication information. The model assumes Parkinson's disease manifests as a small number of progressive states, not specified a priori but discovered in the training data, in which each state is defined by a particular distribution of clinical measures and progression patterns. Insights are drawn both from an analysis of the model's parameters as well as application of the model to discovery and validation cohorts.

(SPECT) scan. Generally healthy, control participants without Parkinson's disease or dementia were also included. All available data, up to 7 years of follow-up, were used in the analysis. 80% of patients with Parkinson's disease were used to train the model (PPMI-training) and the remaining 20% were held-out to serve as an independent test (PPMI-testing); the samples were determined via stratified split to ensure approximate balancing with respect sex, affected side, and Movement Disorder Society-Unified Parkinson's Disease Rating Scale (MDS-UPDRS) score.

PDBP is an aggregation of studies, each with its own inclusion and exclusion criteria.[16] Data are available on the PDBP website. Inclusion criteria in our analysis were a primary diagnosis of Parkinson's disease and two or more study visits that included an MDS-UPRDS assessment. Unlike PPMI, dopaminergic medications are not necessarily an exclusion criterion for enrolment.

The PPMI study protocol was approved by the institutional review board of the University of Rochester (NY, USA) and each PPMI site. The institutional review board of each PDBP site approved the study protocol. Written informed consent was obtained from each study participant in both studies.

Assessments

PPMI collects an extensive array of clinical (motor, non-motor, and medications), imaging, and biomarker assessments. For this analysis, assessments with longitudinal availability in PPMI, and that capture the variety of ways in which symptoms manifest, were selected. These included MDS-UPDRS parts 1, 2, and 3,[17] dopamine transporter SPECT scan specific binding ratios (calculation described previously[14]), activities of daily living score (MSE-ADL), and all measures of autonomic function, cognitive function, impulse control, and sleep or sleepiness (appendix pp 1–2). Dopaminergic medication information was converted to levodopa equivalent daily dose.[18] In some visits, participants were asked to refrain from taking dopaminergic medication, the assessments were done, then the participant took their medication and assessments were re-done (on-off testing). The disease progression model leveraged the on scores, and off scores were used for independent validation.

PDBP data were used to validate our discovery model in an independent dataset. Of the assessments in PPMI, MDS-UPDRS, MSE-ADL, Montreal Cognitive Assessment, and the Epworth Sleepiness Scale were available. Medication information was converted to levodopa equivalent daily dose in the same manner as PPMI. Harmonisation and missing data are discussed in the appendix (pp 4–5).

For **PDBP data** see https://pdbp.ninds.nih.gov/

Seven clinically relevant key outcomes that did not contribute to the discovery model were defined to test the clinical relevance of the identified states (appendix pp 1–2): mild cognitive impairment, dementia, dyskinesias, motor fluctuation, functional impairment from motor fluctuations, Hoehn and Yahr score greater than 3 (cannot ambulate unassisted), and death.

Data pre-processing

The PPMI-training data and healthy control data were analysed with a contrastive latent variable model.[19] This analysis reduces the dimensionality of the participant visits while capturing the major sources of variation in the data. Furthermore, the model separates variation that is shared among Parkinson's disease and healthy control groups (shared space) from variation that is unique to patients with Parkinson's disease (target space). The result of the model is a new set of measures for each participant visit, which are a linear combination of the original clinical observations. This dimensionality reduction retains most of the original information while reducing noise—eg, spurious variation due to exogenous factors such as differences in rater. The resulting model was applied to PPMI-testing and PDBP datasets to obtain the corresponding latent representations.

Modelling

The reduced dimension data were then used for disease progression modelling. Progression modelling was done using a machine learning approach: a personalised input-output hidden Markov model (figure 1).[20] The primary goal of the analysis was to learn a small number of clinically useful states while accounting for medication effects. A state is a discrete label of a patient and has two primary characteristics: a transition model, which describes the probability of changing states, and an observation model, which describes the distribution of clinical measures associated with a state. In this analysis, the transition model was constrained such that a patient can only move progressively (eg, a patient cannot transition from state 3 to state 1); however, patients are not required to progress through all the states (eg, a patient could possibly transition directly from state 1 to state 3). This choice was motivated by the degenerative and heterogeneous nature of Parkinson's disease. We assumed that medications do not modify the underlying disease trajectory;[2] dopaminergic medication effects were incorporated into the observation model and were assumed to be a function of disease state and individualised response. Once the parameters were learned, the model was applied to all visits for a given patient and used to determine the corresponding disease state. In secondary analysis, the model was also used to make predictions about future states. The number of disease states was selected via 5-fold cross validation by maximising a lower bound of the model likelihood using the training data. The interpretation of the disease states and their evolution was enabled by an investigation of the model parameters and visualisations.[21]

Statistical analysis

Several sets of analysis were performed to judge the clinical significance of the learned states and state assignments. First, we interpreted each disease state based on the model parameters. Then, we assessed the agreement of the assigned state with the clinical observation, focusing on clinically relevant measures (eg, tremor, bradykinesia, or rigidity; postural instability or gait difficulty; or non-motor symptoms). We assessed the clinical relevance of the disease states by analysing prevalence of Parkinson's disease outcomes as a function of state assignment. In both cases, the χ^2 test was done for categorical data and the Kruskal-Wallis test was done for ordinal data, and, if significant at $p<0·05$, followed by pairwise testing with correction for multiple testing using the Benjamini-Hochberg procedure.

To evaluate disease trajectories, Kaplan-Meier analysis was applied to analyse time to terminal state as a function of starting state and log-rank test with correction for multiple testing using the Benjamini-Hochberg procedure was used to assess statistical significance.

In the analyses described here, all data contributed by each participant were used to determine state assignment. Additional testing was then done to assess the model's ability to be used in a prospective setting using only baseline visit information to predict assignment of baseline state and whether or not a patient will reach the terminal state in the subsequent 2 years.

The inclusion of medication effects was examined via comparison to a hidden Markov model that does not account for medication effects. The two models were compared using test log-likelihood. The learned medication parameters were compared to observed trends from on-off testing; Kendall's τ was used to assess the magnitude ranking of predicted and observed medication effects on the MDS-UPDRS part 3 (motor subscore).

Role of the funding source

The study was funded by Michael J Fox Foundation for Parkinson's Research (MJFF). Research officers (MF and LAS) at MJFF were involved in study design, data collection, data analysis, data interpretation, and writing of the report.

Results

PPMI data were downloaded July 25, 2018 and medication information was downloaded on Sept 24, 2018. PDBP data were downloaded between June 15 and June 24, 2020. The PPMI-training dataset consisted of 338 participants with Parkinson's disease with 4342 research visits and the PPMI-testing dataset consisted of 85 participants with Parkinson's disease with 1068 research visits. The healthy control cohort included 196 participants with 1268 research visits and PDBP included 610 participants with Parkinson's disease with 2640 visits. A summary of baseline cohort characteristics is shown in the table (baseline ethnicity and race demographic data are shown in the appendix p 3).

Application of the contrastive latent variable model decreased the dimensionality of each visit from 82 measures (appendix pp 1–2) to a 28 measure target space and two measure shared space (appendix pp 9–15).

Combined, this new representation captured 98·6% of the original data's variance. To provide an intuition of the transformed space, we interpret the leading directions of variance (see appendix pp 9–15 for further detail). The four leading directions of variance capture 28%, 9·8%, 7·8%, and 7·5% of the variance, and nominally capture functional impairment, predominantly impacted side of the body, bradykinesia and cognitive measures, and tremor, respectively. The shared space (ie, in which

	Healthy controls (n=196)*	PPMI training (n=338)*	PPMI testing (n=85)*	PPMI testing p value†	PDBP (n=610)‡	PDBP p value†
Number of visits per participant contributing data to analysis	6·5 (1·5)	12·8 (3·0)	12·6 (3·2)	0·45	4·3 (2·8)†	<0·0001
Length of follow-up, years	3·8 (1·7)	5·1 (1·4)	4·9 (1·4)	0·33	3·1 (0·8)†	<0·0001
Sex						
Female	70 (36%)	117 (35%)	29 (34%)	0·97	218 (36%)	0·78
Male	126 (64%)	221 (65%)	56 (66%)	0·97	392 (64%)	0·78
Age at diagnosis, years	··§	61·0 (9·6)	62·0 (9·9)	0·38	59·8 (9·3)¶	0·23
Age at baseline, years	60·8 (11·2)	61·4 (9·6)	62·5 (9·9)	0·36	63·3 (9·1)†	0·0033
Disease duration at baseline visit, years	··§	0·5 (0·5)	0·5 (0·6)	0·49	4·9 (4·5)†¶	<0·0001
Prescribed dopaminergic medication at baseline	··§	0	0	··§	280 (46%)†	<0·0001
LEDD at baseline for patients prescribed dopaminergic medication	··§	··§	··§	··§	229 (164)	··§
Baseline clinical characteristics						
MDS-UPDRS part 1	2·9 (3·0)	5·3 (3·7)	6·5 (5·1)†	0·013	6·7 (5·1)†	<0·0001
MDS-UPDRS part 2	0·5 (1·0)	5·8 (4·1)	6·0 (4·4)	0·70	7·2 (6·0)†	0·0003
MDS-UPDRS part 3	1·2 (2·2)	21·1 (9·1)	20·3 (7·9)	0·46	21·5 (10·2)	0·50
MDS-UPDRS total	4·6 (4·4)	32·2 (13·1)	32·8 (13·2)	0·70	35·5 (16·8)†	0·0023
MSE-ADL	100 (0)	93·2 (5·9)	92·9 (5·9)	0·72	91·5 (10·5)†	0·0055
SCOPA-AUT	5·9 (3·7)	9·4 (6·3)	10·0 (5·6)	0·40	··||	··§
State-train anxiety inventory	57·1 (14·1)	65·0 (17·8)	67·0 (20·5)	0·37	··||	··§
Geriatric Depression Scale	1·3 (2·1)	2·2 (2·3)	2·8 (2·9)†	0·046	··||	··§
Epworth Sleepiness Scale	5·6 (3·4)	5·8 (3·5)	5·9 (3·4)	0·86	7·4 (4·7)†	<0·0001
Rapid eye movement sleep behaviour questionnaire	2·9 (2·3)	4·5 (2·9)	4·6 (2·8)	0·76	··||	··§
QUIP A–E mean	0·3 (0·7)	0·3 (0·7)	0·2 (0·5)	0·18	··||	··§
Montreal Cognitive Assessment	28·2 (1·1)	27·1 (2·3)	27·1 (2·3)	0·85	27·2 (2·7)	0·70
Benton judgement of line orientation	13·1 (2·0)	12·8 (2·2)	12·8 (1·9)	0·72	··||	··§
Semantic fluency test	51·8 (11·2)	48·6 (11·6)	49·1 (11·8)	0·72	··||	··§
Symbol digit modalities	46·8 (10·5)	41·2 (9·8)	41·2 (9·3)	1·0	··||	··§
Hopkins verbal learning test	37·0 (6·1)	34·8 (6·6)	34·9 (7·7)	0·92	··||	··§
Letter number sequencing	10·9 (2·6)	10·6 (2·7)	10·6 (2·6)	0·99	··||	··§
Mean Putamen SBR, mean of left and right	0·1 (0·7)	1·6 (0·7)	1·5 (0·6)	0·60	··||	··§
Mean Caudate SBR, mean of left and right	0·2 (1·0)	3·8 (1·4)	3·7 (1·2)	0·43	··||	··§

Data are mean (SD) or n (%). Baseline ethnicity and race demographic data are shown in the appendix p 3. LEDD=Levodopa equivalent daily dose. MDS-UPDRS=Movement Disorders Society Unified Parkinson's Disease Rating Scale. MSE-ADL=Modified Schwab and England Activities of Daily Living Scale. PDBP=National Institute of Neurological Disorders and Stroke Parkinson's Disease Biomarker Program. PPMI=Parkinson's Progression Markers Initiative. QUIP=Questionnaire for Impulsive-Compulsive Disorders in Parkinson's disease. SBR=specific binding ratio. SCOPA-AUT=Scales for Outcomes in Parkinson's Disease-Autonomic Dysfunction. *PPMI data were download July 25, 2018 and medication information was downloaded on Sept 24, 2018. †p values are calculated as compared to train data using χ^2 test or t test as appropriate; testing was not done for healthy controls. ‡PDBP data were downloaded between June 15 and June 24, 2020. §Not applicable. ¶Data only available for 150 (25%) of 610 PDBP patients. ||Not reported.

Table: **Demographic and baseline clinical characteristics**

the Parkinson's disease and healthy control overlap) captures 3·9% of the variance and is primarily related to sleepiness or sleep disorders, autonomic dysfunction, and anxiety or depression (appendix pp 14–15).

The cross-validation study results in an eight-state model. States were indexed starting from 1, and state 8 is the terminal state (figure 2). The states are primarily differentiated by functional impairment, bradykinesia, and tremor severity but also have differences in gait issues, rigidity, cognitive impairment, anxiety, depression, apathy, sleepiness or sleep disorders, and autonomic dysfunction. State 1 is the mildest and state 8 is the most severe for most measures. Agreement between the model-designated state and the clinical observations for individuals assigned to that state confirms our interpretation (appendix pp 16–17). For instance, states 5 and 8 are associated with greater postural instability or gait difficulty, and most pairwise comparisons of states in these measures as assessed by the MDS-UPDRS are significant at the p less than 0·01 level. The state assignment of individuals developing the seven key clinical outcomes further supports the interpretation:

A

	ADL or functional impairment	Bradykinesia	Tremor	Rigidity	Postural instability and gait issues	Depression, anxiety, or apathy	Cognitive impairment	Sleepiness or sleep disorders	Autonomic dysfunction
State 1	Slight	Mild	Mild	Mild	Slight	Slight	Slight	Slight	Slight
State 2	Mild	Mild	Slight	Mild	Mild	Moderate	Mild	Mild	Mild
State 3	Mild	Moderate	Moderate	Moderate	Mild	Mild	Mild	Mild	Moderate
State 4	Mild	Moderate	Slight	Moderate	Mild	Slight	Mild	Mild	Mild
State 5	Moderate	Severe	Slight	Severe	Moderate	Moderate	Moderate	Moderate	Moderate
State 6	Mild	Moderate	Severe	Moderate	Mild	Mild	Mild	Mild	Mild
State 7	Mild	Moderate	Moderate	Moderate	Mild	Mild	Mild	Mild	Mild
State 8	Severe	Severe	Moderate	Severe	Severe	Severe	Severe	Severe	Severe

B

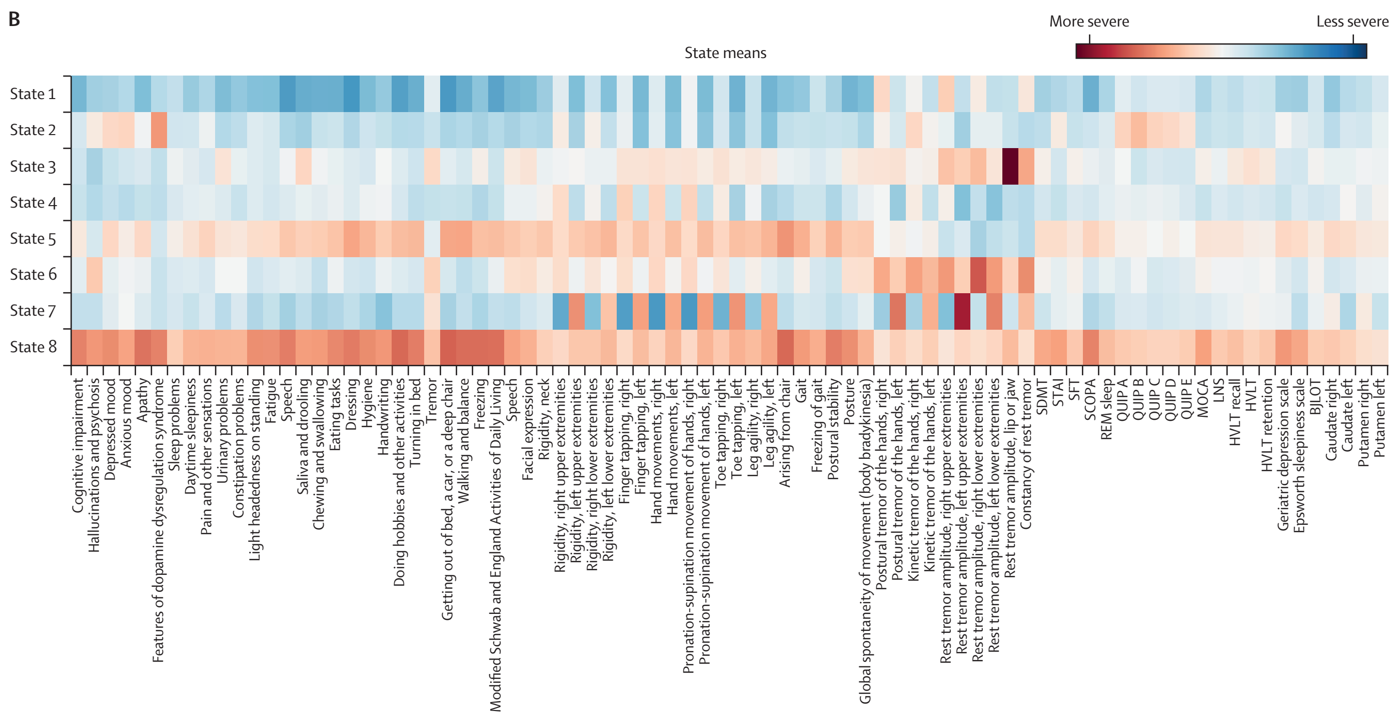


Figure 2: **Qualitative summary of state characteristics (A) and relative values of the 82 clinical measures across the eight states (B)**
All data has been scaled based on the range of values in the Parkinson's Progression Markers Initiative training data sample. Red indicates relatively more severe values and blue indicates relatively less severe values, which might not necessarily mean a larger measure. ADL=Activities of Daily Living. BJLOT=Benton judgement of line orientation. HVLT=Hopkins verbal learning test. LNS=Letter number sequencing. MOCA=Montreal Cognitive Assessment. QUIP=Questionnaire for Impulsive-Compulsive Disorders in Parkinson's Disease. REM=Rapid eye movement. SCOPA=Scales for Outcomes in Parkinson's Disease. SDMT=Symbol digit modalities. SFT=Semantic fluency test. STAI=State-trait anxiety inventory.

state 8 (the terminal state) accounted for the largest proportion of states that had, at any given study visit, the outcomes (appendix pp 18–19). Specifically, in PPMI state 8 accounted for 94 (56%) of 169 mild cognitive impairment, 18 (95%) of 19 dementia, five (71%) of seven dyskinesia, 22 (55%) of 40 motor fluctuations, 58 (68%) of 85 functionally-impactful motor fluctuations, 28 (90%) of 31 Hoehn and Yahr scores greater than three instances, and ten (56%) of 18 deaths. In PDBP, state 8 accounted for nine (82%) of 11 dyskinesia, 13 (54%) of 24 motor fluctuations, 162 (74%) of 219 functionally impactful motor fluctuations, and 65 (97%) of 67 Hoehn and Yahr scores greater than 3 instances. Those in state 8 were significantly more likely to develop dementia or functionally-impairing motor fluctuations compared with all other states; this finding was true for both the PPMI and PDBP datasets.

When compared to a model without medication effects, our model had improved performance on both held-out datasets—ie, data not used in training (figure 3). Overall, bradykinesia and tremor measures were found to improve with medication, whereas non-motor measures were found to be worse with medication (appendix pp 20–21). Application of Kendall's τ to the average predicted effect of medication compared with the medication effect observed in on-off testing data resulted in a statistic of 0·3 for PPMI, indicating adequate consistency in ranking of effect size (appendix pp 22–23).

State 1 was the most observed state at the outset of PPMI with 96 (29%) of 333 participants for the PPMI-training set and 21 (25%) of 83 participants for the PPMI-testing set. The most observed state transitions were from 7 to 8, 6 to 7, and 4 to 6. State transitions directly from 1 to 8, 2 to 3, and 4 to 7 were never observed. 4 (1%) of 333 PPMI-training participants and 3 (4%) of 83 the PPMI-testing participants were in state 8 at the study outset and 138 (41%) of 333 PPMI-training participants and 48 (58%) of 83 PPMI-testing participants reached state 8 by year 5 (appendix pp 24–26). However, Kaplan-Meier analysis revealed that the ranking of the starting state did not match the ranking of reaching state 8 within 5 years (figure 4). Participants starting in state 2 as compared with state 1 had a hazard ratio (HR) of 6·4 (95% CI 3·2–12·6) for reaching state 8 at year 5, whereas compared with state 1 state 3 has a HR of 1·9 (1·3–2·7) and state 4 has a HR of 1·4 (1·1–1·8). Overall, participants in state 5 had the shortest time to state 8 (median time of 2·75 [95 CI% 1·75–4·25] years for the PPMI-training set and 2·25 [0·5–3·25] years for the PPMI-testing set) and were significantly more likely to reach state 8 compared with participants starting in states 1, 3, and 4 (figure 4B).

Results of our analysis confirm that the discovered model can also be applied in a prospective setting. For 55 of the 83 PPMI-testing participants, baseline state assignment was consistent whether using only cross-sectional data or all available patient data. The most common discrepancy was assigning state 1 to 4 and vice versa, for eight of the inconsistently classified participants. Analogously for 62 of the 83 PPMI-testing participants, the binary terminal state assignment prediction was consistent. For additional details, see the appendix (pp 27–29).

The observed trajectory trends were confirmed in the validation cohort. Based on baseline demographics, the PDBP cohort was slightly more progressed and had longer disease duration at baseline (table). In PDBP, state 4 was the most observed at outset with 167 (27%) of 610 participants in PDBP as compared with 66 (20%) of 33 participants for the PPMI-training dataset and 18 (22%) of 83 participants for the PPMI-testing dataset. The most observed state transitions were from 4 to 5, 5 to 8, and 4 to 6. 51 (8%) of 610 participants are in state 8 at study outset and 156 (26%) of 610 participants reached state eight by year 3. Kaplan-Meier analysis confirmed the trends observed in PPMI (figure 4). The HR of reaching the terminal state (state 8) when starting in state 2 versus state 1 is 18·4 (95%CI 6·6–51·1) at year 3 whereas state 3 versus state 1 HR 1·6 (0·9–2·8) and state 4 versus state 1 HR 1·5 (1·1–2·0). Median time to state 8 starting from state 5 was 2·25 years (95% CI 1·25–not reached). 464 (76%) of the 610 PDBP participants had consistent baseline state assignment whether using only cross-sectional data or all available patient data. Again, the most common discrepancy was between states 1 and 4 (37 inconsistent assignments). For the binary terminal state prediction, 496 (81%) of the 610 participants had consistent predictions.

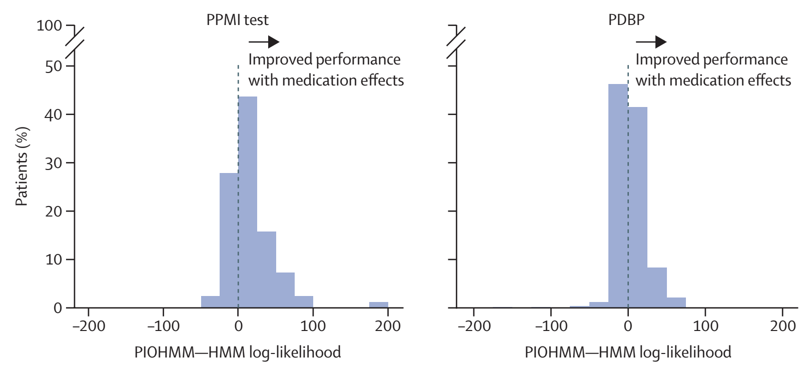


Figure 3: **Differences in the log-likelihood of the models with (personalised input-output hidden Markov model) and without (hidden Markov model) personalised and medication effects for each patient in the PPMI-test and PDBP datasets**

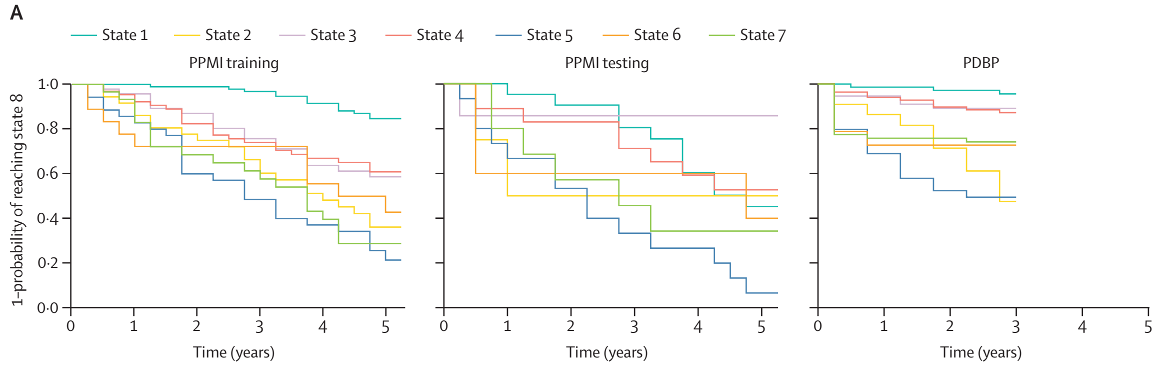
Most patients have a higher likelihood under the proposed PIOHMM model with 58 (70%) of 83 patients having a higher likelihood under the PIOHMM model for PPMI testing and 317 (52%) of 610 patients having a higher likelihood under the PIOHMM model for PDBP. PPMI testing average likelihoods are –346 (SD 120) for the HMM and –330 (105) for the PIOHMM and PDBP average likelihoods are –127 (114) for the HMM and –123 (107) for the PIOHMM (a larger likelihood implies a better fitting model). HMM=hidden Markov model. PDBP=National Institute of Neurological Disorders and Stroke Parkinson's Disease Biomarker Program. PIOHMM=personalised input-output hidden Markov model. PPMI=Parkinson's Progression Markers Initiative.

Discussion

With a novel machine-learning approach and using data that can be collected during routine clinical visits, we identified eight Parkinson's disease states, which manifest differently in terms of their motor and non-motor symptoms and their progression. Severe outcomes were associated with the terminal Parkinson's disease state (state 8). Importantly, we were able to validate our findings in an independent Parkinson's disease cohort. Our model accounts for the heterogeneity of Parkinson's disease in its manifestations, progression, and effect of medications, and offers insights into Parkinson's disease and how it progresses.

Parkinson's disease is a heterogeneous disorder, with marked inter-rater and intra-rater variability in its manifestations, especially early in the disease course. Our application of contrastive latent modelling allowed for the removal of population-level variation in a data-driven manner, which has been noted in past studies as potentially masking important relationships.[7] Our results suggest that if classic dimensionality reduction methods such as principal component analysis are used, as is often done,[17,22] the variability of autonomic function, sleepiness or sleep disorders, depression, and anxiety might be overstated as these measures also vary in an

A



B

PPMI training	2	3	4	5	6	7
1	<0·0001	0·00019	0·00019	<0·0001	<0·0001	<0·0001
2	..	0·11	0·098	0·28	0·71	0·60
3	..	..	0·91	0·0017	0·32	0·018
4	..	..	..	0·0004	0·18	0·0061
5	..	..	..	..	0·18	0·59
6	..	..	..	..	..	0·37

PPMI testing	2	3	4	5	6	7
1	0·89	0·37	0·92	0·0075	0·89	0·37
2	..	0·56	0·91	0·56	0·91	0·91
3	..	..	0·22	0·0055	0·21	0·12
4	..	..	..	0·0069	0·61	0·33
5	..	..	..	..	0·18	0·22
6	..	..	..	..	..	0·96

PDBP	2	3	4	5	6	7
1	<0·0001	0·072	0·011	<0·0001	<0·0001	<0·0001
2	..	0·00076	<0·0001	0·53	0·25	0·15
3	..	..	0·75	<0·0001	0·046	0·046
4	..	..	..	<0·0001	0·018	0·013
5	..	..	..	..	0·054	0·0096
6	..	..	..	..	..	0·88

***Figure 4:* Kaplan-Meier analysis of time to state 8 from each of the other seven starting states for the three datasets over a period of 5 years in PPMI and 3 years in PDBP (A) and a log-rank test to test the statistical significance of time to state 8 for all pairs (B)**

(A) The ranking of the states at 5 years does not match the initial state ranking. Notably, states 2 and 5 have a higher probability of reaching state 8, particularly when compared with states 1, 3, and 4. (B) Grey shaded cells indicate statistically significant pairs at p values less than 0·05 after correction for multiple testing using the Benjamini-Hochberg procedure. PDBP=National Institute of Neurological Disorders and Stroke Parkinson's Disease Biomarker Program. PPMI=Parkinson's Progression Markers Initiative.

age-matched control. Accounting for this variation is an important step in discovering robust disease states.

Most work in data-driven understanding of Parkinson's disease has focused on the discovery of subtypes, using baseline cross-sectional data and clustering techniques. In defining subtypes, an assumption is made that there are unique (deterministic) progression patterns of Parkinson's disease within a subtype,[7–11,23–26] and consequently the applied model constraints reflect this. However, the utility of Parkinson's disease subtyping is an open area of research,[27] and the idea of a continuum of Parkinson's disease remains viable, specifically that "PD [Parkinson's disease] subtypes are unlikely to be distinct, non-overlapping entities but are more likely to represent typical phenotypes within a multidimensional spectrum resulting from variable contributions of a number of simultaneous pathological processes."[28] This hypothesis is supported by the observation that Parkinson's disease subtypes show instability, especially early in the disease[13,29] and that advanced Parkinson's disease is clinically very similar despite heterogeneity in early disease.[11,30] In our model setup, participants could follow any progression path allowable under the progressive constraint, including sequential progression, non-overlapping pathways (ie, subtypes), and non-sequential, overlapping pathways. Therefore, our approach is more flexible compared with previously published models in terms of possible progression pathways. Furthermore, data collected up to 7 years in an early Parkinson's disease cohort was used to discover Parkinson's disease states and their corresponding progression, thus accounting for the heterogeneity of Parkinson's disease that might not be reflected in models that use only cross-sectional data. Our findings indicate that our model discovered non-sequential, overlapping disease progression trajectories, supporting the use of non-deterministic disease progression models, and suggesting static subtype assignment might be ineffective at capturing the full spectrum of Parkinson's disease progression. Interestingly, state 2 as well as state 5 had the mildest tremor of all states but also comparatively rapid progression to the terminal state, and several studies[4,8] have shown that absence of tremor is a poor prognostic sign in Parkinson's disease.

Our model accounted for effects of Parkinson's disease medications. Dopaminergic medications have several desirable effects such as improving bradykinesia and rigidity, but also adverse effects such as hypersomnolence and impulse-control disorders.[2] As with other aspects of Parkinson's disease, there might be marked intra-individual and inter-individual variability for the effect of Parkinson's disease medications on both these aspects.[31] Although extensive work has previously been done to apply machine-learning techniques for modelling of progression in early Parkinson's disease,[7–11,23–26] previous models have limited or no accounting of Parkinson's disease medication effects. By contrast, we modelled medication effects as deviations from disease states, which vary across states and participants. An ideal model might allow these factors to vary together, with individualised response varying over time. However, most studies do not have enough data to accurately estimate time-varying, individualised response. Therefore, we compromise by allowing each patient to have a non-time-varying personalised response to medication and a time-varying response which is a function of disease state.

There are several limitations to our study. Because PPMI enrolled a cohort of de-novo Parkinson's disease, late-stage disease might be underrepresented. Our use of up to 7 years' worth of data to develop our model, as well as its validation in the more advanced PDBP cohort, helps mitigate this concern. Because both PPMI and PDBP include individuals who can visit tertiary care research centres, and because the majority of the cohort identify as White, our results might not be generalisable to the entire Parkinson's disease population.

In summary, we developed a statistical progression model of early Parkinson's disease that accounts for intra-individual and inter-individual variability and medication effects, circumventing limitations of previous models that have not accounted for these important factors. Our model discovered non-sequential, overlapping disease progression trajectories, supporting the use of non-deterministic disease progression models, and suggesting static subtype assignment might be ineffective at capturing the full spectrum of Parkinson's disease progression. Our findings support the hypothesis of heterogeneous progression and has implications for patient stratification and management. With validation of the model in real-world settings with diverse cohorts capturing a range of clinically relevant short-term and long-term outcomes, and further refinement to leverage a minimal set of emerging clinical or biomarker assessments that best identify granular states of Parkinson's disease progression, our model has the potential to translate to a prognostication tool in the clinic, and a predictive tool that might be used for clinical trial sample enrichment.

Contributors

SG and JH had the idea for the study, which was designed with input from MF, LMC, KAS, and LAS. All authors had access to the data, which was verified by KAS and MD. SG and KAS developed methods for analysing the data, and KAS, MD, and SG did the data analysis. All authors contributed to the analysis. KAS and LMC wrote the first draft of the manuscript and all authors contributed to writing the report. All authors approved the final version and were involved in the decision to submit for publication.

Declaration of interests

JH, KAS, KN, and SG are employees of IBM Research. MD is a former employee of IBM Research and completed the work while employed at IBM Research. MF is an employee of the Michael J Fox Foundation. LAS is a former employee of the Michael J Fox Foundation and completed the work while employed at the Michael J Fox Foundation. LMC is an employee of the University of Pittsburgh.

Data sharing

Data used in this Article are publicly available from were obtained from the Parkinson's Progression Markers Initiative database (www.ppmi-info.org/data) and the National Institute of Neurological Disorders and Stroke Parkinson's Disease Biomarker Program (https://pdbp.ninds.nih.gov). The contrastive latent variable model code is available via https://github.com/kseverso/contrastive-LVM. The personalised input-output hidden Markov models code is available via https://github.com/kseverso/DiseaseProgressionModeling-HMM. Code showing the data pre-processing and modelling and analysis pipelines is available via https://github.com/kseverso/Discovery-of-PD-States-using-ML.

Acknowledgments

This study was funded by the Michael J Fox Foundation. The Parkinson's Progression Markers Initiative is a public-private partnership and is funded by the Michael J Fox Foundation for Parkinson's Research funding partners, including AbbVie, Allergan, Amathus Therapeutics, Avid Radiopharmaceuticals, Biogen, BioLegend, Bristol Myers Squibb, Celgene, Denali, GE Healthcare, Genentech, GlaxoSmithKline, Handl Therapeutics, Insitro, Janssen Neuroscience, Lilly, Lundbeck, Merck, Meso Scale Discovery, Pfizer, Piramal, Prevail, Roche, Sanofi Genzyme, Servier, Takeda, Teva, UCB, Verily, and Voyager Therapeutics (Golub Capital). Data and biospecimens used in preparation of this manuscript were obtained from the Parkinson's Disease Biomarkers Program (PDBP) Consortium, supported by the National Institute of Neurological Disorders and Stroke at the National Institutes of Health. Investigators include: Roger Albin, Roy Alcalay, Alberto Ascherio, Thomas Beach, Sarah Berman, Bradley Boeve, F DuBois Bowman, Shu Chen, Alice Chen-Plotkin, William Dauer, Ted Dawson, Paula Desplats, Richard Dewey, Ray Dorsey, Jori Fleisher, Kirk Frey, Douglas Galasko, James Galvin, Dwight German, Lawrence Honig, Xuemei Huang, David Irwin, Kejal Kantarci, Anumantha Kanthasamy, Daniel Kaufer, James Leverenz, Carol Lippa, Irene Litvan, Oscar Lopez, Jian Ma, Lara Mangravite, Karen Marder, Laurie Orzelius, Vladislav Petyuk, Judith Potashkin, Liana Rosenthal, Rachel Saunders-Pullman, Clemens Scherzer, Michael Schwarzschild, Tanya Simuni, Andrew Singleton, David Standaert, Debby Tsuang, David Vaillancourt, David Walt, Andrew West, Cyrus Zabetian, Jing Zhang, and Wenquan Zou. The PDBP Investigators have not participated in reviewing the data analysis or content of the manuscript.

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